

BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

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Due Date

Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `~/BEE 4750 /HW 3/hw3-parker`
Precompiling project...
 466.1 ms   WorkerUtilities
 472.1 ms   IteratorInterfaceExtensions
 471.7 ms   TensorCore
```

536.2 ms	LaTeXStrings
469.3 ms	Contour
478.7 ms	StatsAPI
356.6 ms	Measures
918.2 ms	Format
479.9 ms	Statistics
503.9 ms	OrderedCollections
653.6 ms	Grisu
302.9 ms	DataValueInterfaces
786.8 ms	FillArrays
443.3 ms	Requires
360.2 ms	StableRNGs
443.3 ms	Unzip
310.8 ms	Reexport
502.0 ms	DocStringExtensions
378.1 ms	SimpleBufferStream
396.2 ms	PtrArrays
612.3 ms	InlineStrings
612.1 ms	URIs
1750.5 ms	MacroTools
411.4 ms	InvertedIndices
1029.5 ms	IrrationalConstants
499.4 ms	NaNMath
475.3 ms	DelimitedFiles
616.1 ms	TranscodingStreams
421.0 ms	DataAPI
355.2 ms	BitFlags
376.4 ms	Scratch
720.3 ms	ConcurrentUtilities
982.5 ms	Crayons
598.2 ms	LoggingExtras
904.4 ms	MbedTLS
1071.9 ms	SentinelArrays
754.2 ms	Preferences
624.8 ms	Compat
579.7 ms	ExceptionUnwrapping
1521.8 ms	PDMats
613.5 ms	TableTraits
704.5 ms	Statistics → SparseArraysExt
401.9 ms	Showoff
1210.5 ms	UnicodeFun
671.3 ms	FillArrays → FillArraysStatisticsExt
637.4 ms	FillArrays → FillArraysSparseArraysExt

567.4 ms	AliasTables
791.2 ms	CodecZlib
1826.4 ms	FixedPointNumbers
1596.3 ms	DataStructures
868.4 ms	Missings
935.2 ms	PooledArrays
1450.9 ms	LogExpFunctions
958.0 ms	RelocatableFolders
750.2 ms	PrecompileTools
767.0 ms	JLLWrappers
886.9 ms	Compat → CompatLinearAlgebraExt
1105.7 ms	FillArrays → FillArraysPDMatsExt
1338.0 ms	SortingAlgorithms
1431.5 ms	QuadGK
1776.2 ms	ColorTypes
2331.3 ms	Tables
742.6 ms	OpenSSL_jll
529.5 ms	Graphite2_jll
532.9 ms	Libmount_jll
2046.7 ms	RecipesBase
1927.2 ms	StringManipulation
435.2 ms	EpollShim_jll
412.7 ms	LLVMOpenMP_jll
432.3 ms	Bzip2_jll
391.7 ms	Xorg_libICE_jll
430.6 ms	Rmath_jll
397.4 ms	Xorg_libXau_jll
416.3 ms	libfdk_aac_jll
421.7 ms	libpng_jll
438.8 ms	LAME_jll
397.3 ms	fzf_jll
406.7 ms	LERC_jll
507.5 ms	JpegTurbo_jll
472.8 ms	Ogg_jll
508.2 ms	XZ_jll
365.1 ms	mtdev_jll
382.9 ms	Xorg_libXdmcp_jll
419.3 ms	x265_jll
455.0 ms	x264_jll
441.8 ms	Zstd_jll
455.9 ms	libaom_jll
434.1 ms	Expat_jll
424.8 ms	LZO_jll

440.0 ms	Opus_jll
403.7 ms	Xorg_xtrans_jll
386.6 ms	libevdev_jll
421.6 ms	Libiconv_jll
414.1 ms	Libffi_jll
412.5 ms	eudev_jll
390.3 ms	Libuuid_jll
658.7 ms	OpenSpecFun_jll
416.4 ms	FriBidi_jll
588.2 ms	FilePathsBase
548.8 ms	ColorTypes → StyledStringsExt
1305.8 ms	Pixman_jll
2697.8 ms	ColorVectorSpace
445.8 ms	Xorg_libSM_jll
1275.4 ms	FreeType2_jll
3301.6 ms	OpenSSL
1080.2 ms	Rmath
1350.9 ms	JLFzf
4577.8 ms	Colors
4931.8 ms	StatsBase
1324.6 ms	Ghostscript_jll
720.4 ms	libvorbis_jll
358.8 ms	Xorg_libxcb_jll
8681.3 ms	Parsers
380.1 ms	DBus_jll
373.2 ms	Wayland_jll
350.9 ms	libinput_jll
876.7 ms	GettextRuntime_jll
924.3 ms	FilePathsBase → FilePathsBaseMmapExt
1281.5 ms	FilePathsBase → FilePathsBaseTestExt
1828.1 ms	Libtiff_jll
311.3 ms	Xorg_xcb_util_jll
1902.6 ms	SpecialFunctions
307.1 ms	Xorg_libX11_jll
2611.8 ms	Fontconfig_jll
1728.4 ms	JSON
316.3 ms	Xorg_xcb_util_image_jll
1792.6 ms	InlineStrings → ParsersExt
363.8 ms	Xorg_xcb_util_keysyms_jll
370.3 ms	Xorg_xcb_util_renderutil_jll
441.5 ms	Xorg_xcb_util_wm_jll
2041.7 ms	Glib_jll
509.0 ms	ColorVectorSpace → SpecialFunctionsExt

411.5 ms	Xorg_libXrender_jll
1155.3 ms	HypergeometricFunctions
4498.8 ms	ColorSchemes
389.4 ms	Xorg_libXext_jll
474.1 ms	Xorg_libXfixes_jll
422.3 ms	Xorg_libxkbfile_jll
501.1 ms	Xorg_xcb_util_cursor_jll
819.8 ms	WeakRefStrings
4950.5 ms	Latexify
453.8 ms	Xorg_libXinerama_jll
400.1 ms	Xorg_libXrandr_jll
376.6 ms	Libglvnd_jll
360.8 ms	Xorg_libXi_jll
368.8 ms	Xorg_libXcursor_jll
407.4 ms	Xorg_xkbcomp_jll
2207.4 ms	StatsFuns
1296.5 ms	Latexify → SparseArraysExt
1099.6 ms	Xorg_xkeyboard_config_jll
2962.5 ms	Cairo_jll
506.9 ms	xkbcommon_jll
565.2 ms	Vulkan_Loader_jll
1517.2 ms	HarfBuzz_jll
1971.9 ms	libass_jll
846.6 ms	Pango_jll
5234.3 ms	Qt6Base_jll
9496.1 ms	PlotUtils
465.5 ms	libdecor_jll
1608.7 ms	Qt6ShaderTools_jll
3645.9 ms	FFMPEG_jll
9047.8 ms	Distributions
566.3 ms	GLFW_jll
3991.1 ms	PlotThemes
1767.3 ms	FFMPEG
2572.4 ms	Qt6Declarative_jll
20109.4 ms	HTTP
588.2 ms	Qt6Wayland_jll
1508.2 ms	Distributions → DistributionsTestExt
1629.5 ms	GR_jll
4958.7 ms	RecipesPipeline
16889.5 ms	CSV
28853.5 ms	PrettyTables
2785.7 ms	GR
29396.6 ms	DataFrames

```
1600.2 ms    Latexify → DataFramesExt
40598.2 ms   Plots
177 dependencies successfully precompiled in 81 seconds. 37 already precompiled.
```

```
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day⁻¹, and the decay rates of CBOD and NBOD are 0.35 and 0.25 day⁻¹, respectively. The river's saturated dissolved oxygen concentration is 10 mg/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance d km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m ³ /d	10,000 m ³ /d	15,000 m ³ /d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

$$K_a = 0.55 \text{ per day}$$

$$K_c = 0.35 \text{ per day}$$

$$K_n = 0.25 \text{ per day}$$

$$C_s = 10 \text{ mg/L}$$

Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

```
u = 6 #km/day
dist = 15 #km
r = 0.55 #1/day
kc = 0.35 #1/day
kn = 0.25 #1/day
Cs = 10 #mg/L

Q_r0 = 100000 #m3/day
DO_r0 = 7.5 #mg/L
C_r0 = 5 #mg/L
N_r0 = 5 #mg/L

Q_w1 = 10000 #m3/day
DO_w1 = 5 #mg/L
C_w1 = 50 #mg/L
N_w1 = 35 #mg/L

Q_w2 = 15000 #m3/day
DO_w2 = 5 #mg/L
C_w2 = 45 #mg/L
N_w2 = 35 #mg/L

Q_out1 = Q_r0 + Q_w1

function conc_initial(c0, q0, c1, q1, qout)
    ((c0*q0) + (c1*q1)) / qout
end

DO_out1 = conc_initial(DO_r0, Q_r0, DO_w1, Q_w1, Q_out1)
C_out1 = conc_initial(C_r0, Q_r0, C_w1, Q_w1, Q_out1)
N_out1 = conc_initial(N_r0, Q_r0, N_w1, Q_w1, Q_out1)

function conc(x, u, r, kc, kn, D00, C0, N0, Cs)
    a1 = exp(-r*x/u)
    a2 = (kc/(r-kc))*(exp(-kc*x/u) - a1)
    a3 = (kn/(r-kn))*(exp(-kn*x/u) - a1)
    Cs*(1 - a1) + D00*a1 - C0*a2 - N0*a3
```

```

end

# UNTIL X = 15 RIGHT BEFORE SECOND WASTE STREAM INPUT
x1 = range(0.0, stop=15.0, length=1501)      # <- use 0..15 here
C1 = conc.(x1, u, r, kc, kn, DO_out1, C_out1, N_out1, Cs)

# values at x=15
# (L0 and N0 were undefined; you meant the mixed CBOD/NBOD after the first discharge)
L15 = C_out1 * exp(-kc*15.0/u)                # scalar math is fine
N15 = N_out1 * exp(-kn*15.0/u)
C15 = C1[end]                                # indexing uses [end], not C1(end)

# SECOND WASTE STREAM INPUT AT X = 15
Q_out2 = Q_out1 + Q_w2
DO_out2 = conc_initial(C15, Q_out1, DO_w2, Q_w2, Q_out2)
C_out2 = conc_initial(L15, Q_out1, C_w2, Q_w2, Q_out2)
N_out2 = conc_initial(N15, Q_out1, N_w2, Q_w2, Q_out2)

x2 = range(15.0, stop=50.0, length=3501)
C2 = conc.(x2 .- 15.0, u, r, kc, kn, DO_out2, C_out2, N_out2, Cs)

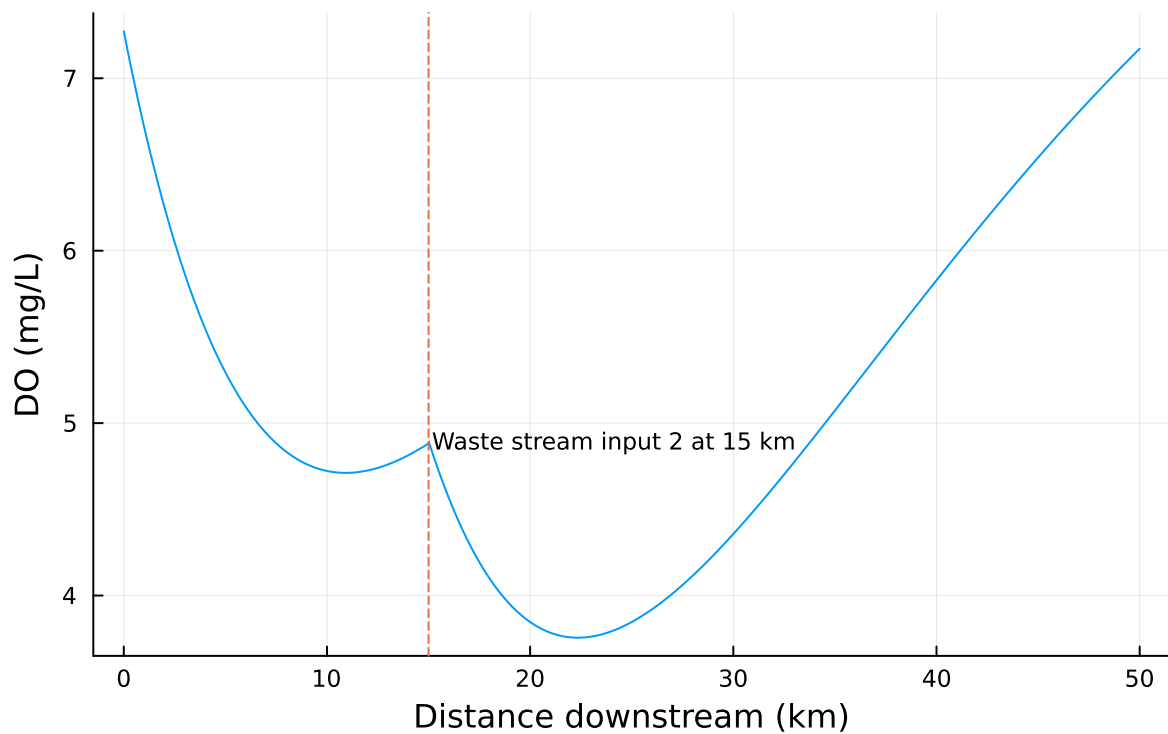
x_total = [x1; x2[2:end]]
C_total = [C1; C2[2:end]]
imin = argmin(C_total)
minDO = C_total[imin]
x_at_min = x_total[imin]
println("Minimum DO = $(round(minDO, digits=3)) mg/L at x = $(round(x_at_min, digits=3)) km")

# plot
plot(x_total, C_total, xlabel = "Distance downstream (km)", ylabel = "DO (mg/L)",
      title = "DO Concentration Downstream with Two Waste Inputs", legend = false)
vline!([15.0], linestyle = :dash)
annotate!(15.2, DO_out2, text("Waste stream input 2 at 15 km", :left, 8))

```

Minimum DO = 3.756 mg/L at x = 22.35 km

DO Concentration Downstream with Two Waste Inputs



Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

1.2 Solution

```
dx = 0.5 #km
L = 50 #km

function do_numerical(L, dx, C0, B0, N0, ka, kn, kc, Cs, U) #Numerical implementation function
    n = Int(L / dx) # number of length steps
    C = zeros(n + 1)
    B = zeros(n + 1)
    N = zeros(n + 1)
    C[1] = C0
```

```

    B[1] = B0
    N[1] = N0
    for i = 1:n
        B[i+1] = B[i] * exp(-kc * dx / U)
        N[i+1] = N[i] * exp(-kn * dx / U)
        C[i+1] = C[i] + (dx / U) * (ka * (Cs - C[i]) - kc * B[i] - kn * N[i])
    end
    return (C, B, N)
end

#Right after first waste input
C1, B1, N1 = do_numerical(15.0, dx, DO_out1, C_out1, N_out1, r, kn, kc, Cs, u)

Q_out2 = Q_out1 + Q_w2
DO_15 = C1[end]
L_15 = B1[end]
N_15 = N1[end]

DO_mix2 = conc_initial(DO_15, Q_out1, DO_w2, Q_w2, Q_out2)
L_mix2 = conc_initial(L_15, Q_out1, C_w2, Q_w2, Q_out2)
N_mix2 = conc_initial(N_15, Q_out1, N_w2, Q_w2, Q_out2)

#Right after second waste input
C2, B2, N2 = do_numerical(35.0, dx, DO_mix2, L_mix2, N_mix2, r, kn, kc, Cs, u)

x = [0.0:dx:15.0; 15.0+dx:dx:50.0]
DO_tot = [C1; C2[2:end]]

imin_num = argmin(DO_tot)
println("Numerical min DO = $(round(DO_tot[imin_num], digits=3)) mg/L at x = $(round(x[imin_num], digits=3)) km")

# Plot on same axes as 1.1 (if you computed x_all, C_all); otherwise just plot numerical.
if @isdefined(x_total) && @isdefined(C_total)
    plot(x_total, C_total, label="Analytic", xlabel="Distance downstream (km)", ylabel="DO (mg/L)",
         title="DO Concentration: Analytic vs Numerical (dx = 0.5 km)")
    plot!(x, DO_tot, linestyle=:dash, label="Numerical (dx = 0.5 km)")
    vline!([15.0], linestyle=:dash, label="Discharge #2 (15 km)")
else
    plot(x, DO_tot, label="Numerical (dx = 0.5 km)", xlabel="Distance downstream (km)", ylabel="DO (mg/L)",
         title="DO Concentration (Numerical, lecture scheme)")
end

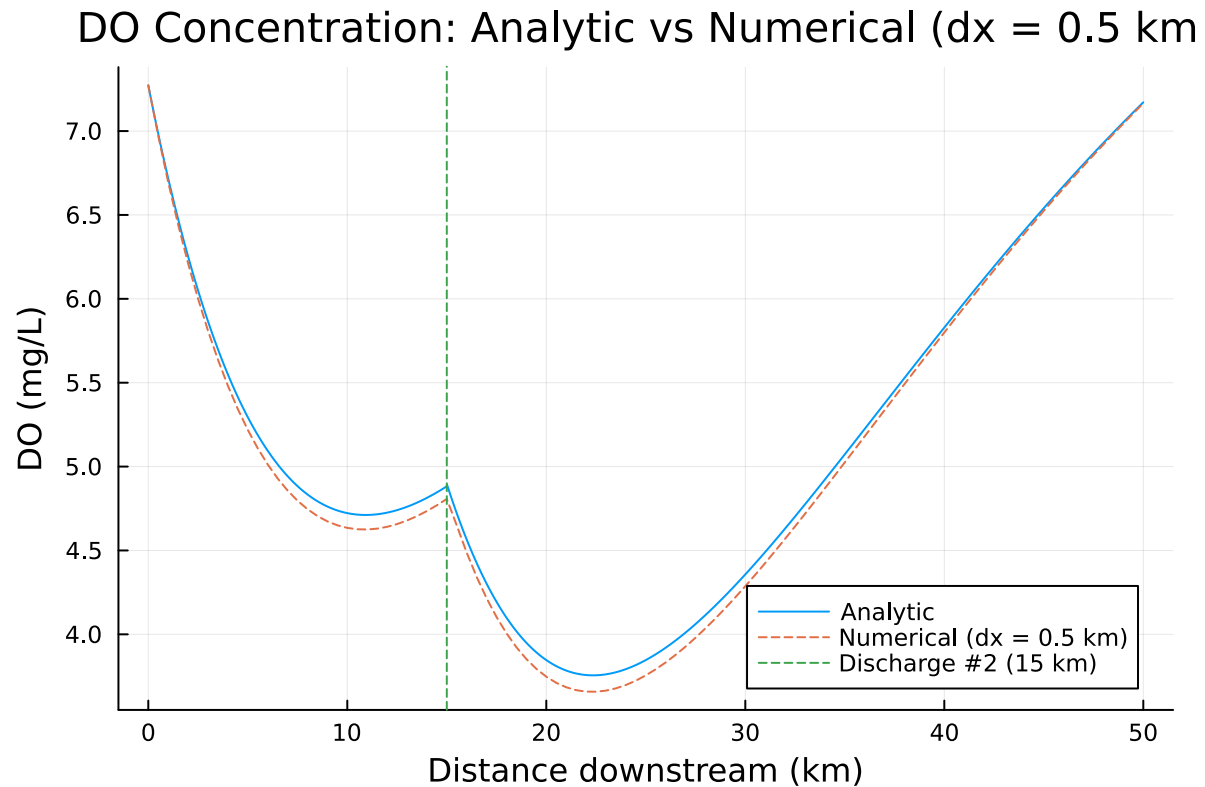
```

```

vline!([15.0], linestyle=:dash, label="Discharge #2 (15 km)")
end

```

Numerical min DO = 3.658 mg/L at x = 22.5 km



Explanation: Through the numerically-integrated solution the minimum value is 3.658 mg/L of DO at x = 22.5 km, compared to 3.756 mg/L of DO at x = 22.35 km using the analytical solution. So, the minimum value has decreased with the numerical solution. This could be because of numerical errors, such as truncation error. This is because the Euler method can over or under-estimate oxygen-defecit with each step depending on the line's curvature since the area under the curve is approximated as a rectangle. This causes the minimum to differ from the analytic solution.

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the

4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal(2.0, 0.15) distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

References

List any external references consulted, including classmates.